ANANYO SEN

MSc in Data Science and AI, RKMVERI, Belur

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GitHub
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SUMMARY

My experience combines a strong foundation in machine learning and deep learning with hands-on experience in building end-to-end AI and data-driven systems. My mathematical background further enables me to explore problems at their core, understanding the underlying concepts and processes.

EDUCATION

Ramakrishna Mission Vidyamandira, Belur
BSc (Honours) in Mathematics, 2022–25
CGPA –6.91

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science and Artificial Intelligence, 2025–27
CGPA – 6.65(till Sem 2)

RESEARCH PROJECTS

Properties of Bessel Function

RKM Vidyamandira Jan 2025 – Apr 2025
Explored the theory and applications of Bessel functions, including recurrence relations, generating functions, orthogonality, and Fourier-Bessel expansions, with applications to solving Laplace's equation in cylindrical coordinates.

SKILLS

- **Languages:** Python, JavaScript, R Programming, LaTeX, C++
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, Matplotlib
- **OS:** Windows, Ubuntu, Linux, MacOS
- **Tools:** Git/GitHub, MS Office

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning
- Probability and Statistics, Linear Algebra, Time Series Analysis, Graph Theory
- Distributed Computing, Data Structures and Algorithms

ACHIEVEMENTS

- JBNSTS Junior Scholarship by the Government of West Bengal in 2020
- 1st place in a College Fest event, Enigma Quest, Puzzle Solving from Perceptron'25 at RKMVERI
- 2nd place in a Poster Presentation on the topic Real Analysis And Its Applications at RKM Vidyamandira.

Certificates

- NPTEL – Programming in Modern C++ (2025)
- NPTEL – Generative AI for Computer Vision (2026)
- NPTEL – Introduction to Large Language Models (2026)

PROJECTS

Distributed Gaming Analytics System Using Ray

RKMVERI Feb 2026 – May 2026
A scalable gaming analytics platform that leverages distributed computing with Ray to perform parallel player analysis and prediction in real time. In this project, prediction using the random forest regressor and analysis using statistical analysis are done, with a web app launched using Flask UI.

Event-Driven Multi-Modal Sports Video Understanding

RKMVERI Feb 2026 – Apr 2026
Sports video understanding is a challenging task due to dynamic scenes, temporal dependencies, and interactions among multiple entities.
The objective of this project was to develop a deep learning framework capable of detecting important events, modeling tactical interactions, and estimating match intensity for automated sports analysis and future commentary generation.

Deep Q-Networks for Partially Observable Treasure Hunt

RKMVERI Oct 2025 – Dec 2025
Reinforcement Learning agents often struggle in environments with limited visibility, dynamic obstacles, and sparse rewards. The objective of this project was to develop an intelligent agent capable of navigating a partially observable treasure hunt environment where the agent collects treasures while avoiding static and dynamic traps.
The environment was designed with multiple difficulty levels and adaptive game elements to investigate the ability of Deep Reinforcement Learning methods to learn efficient navigation policies under uncertainty.

Prediction of Video Game Sales

RKMVERI Sep 2025 – Nov 2025
Predicting video game sales is important for developers and publishers to support marketing strategies, budgeting, and revenue forecasting. This project performs exploratory data analysis, feature engineering, regression modeling, evaluation, and visualization to predict global sales.

RESEARCH EXPERIENCE

Statistical Methods and Literature Review for Machine Learning in Mental Health

RKMVERI May 2026 – Jul 2026
Conducted a structured study of statistical hypothesis testing, ANOVA, power analysis, sample size estimation, and non-parametric methods, alongside a comprehensive literature review on machine learning applications in mental health. Analyzed systematic and primary research papers, reviewed psychometric validation studies (SERATS, SERAPTS, FERATS), and presented technical findings through academic seminars.

Computational Analysis of Comic Narrative Flow

Flame University May 2026 – Sep 2026
Conducted a literature review on computational comic understanding, studying machine learning approaches for modeling narrative flow. Analyzed research on emotion recognition, panel transition understanding, multimodal reasoning, and context inference to understand how AI interprets comic sequences and identify potential research directions.